Hosein Hadipour

Curriculum Vitae

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Google Scholar
DBLP
IACR Profile



Research Profile

I am a postdoctoral researcher in symmetric-key cryptography at Ruhr University Bochum, with a Ph.D. in Computer Science from TU Graz (2024). My research interests span the design and security analysis of cryptographic primitives, including theoretical and practical cryptanalysis and the development of automated methods and tools for evaluating their security. I have published at CRYPTO and EUROCRYPT, as well as in ToSC and TCHES.

Research Appointments

Feb. 2025–Present **Postdoctoral Researcher**, *Ruhr University Bochum*, Bochum, Germany.
Conducting research on symmetric-key cryptanalysis under the supervision of Prof. Gregor Leander in the Symmetric Cryptography Group.

2022–2024 **Ph.D. Candidate**, *Graz University of Technology*, Graz, Austria.
Conducted research on symmetric-key cryptanalysis under the supervision of Dr. Maria Eichlseder. Institute profile.

2021–2022 **Ph.D. Student**, *Graz University of Technology*, Graz, Austria.
Conducted research on symmetric-key cryptanalysis under the supervision of Dr. Maria Eichlseder in the Cryptography Research Area.

Education

2021–2024 **Ph.D. in Computer Science**, *Graz University of Technology*, Graz, Austria.

- Thesis: Automated Methods in Design and Analysis of Symmetric-Key Primitives
- Supervisor: Dr. Maria Eichlseder

2014–2016 **Master of Pure Mathematics**, *University of Tehran*, Tehran, Iran.

- Thesis: Topics in Algebraic Attacks on Cryptosystems
- Supervisor: Dr. Hosein Sabzrou

2012–2016 **Bachelor of Electrical Engineering**, *K. N. Toosi University of Technology*, Tehran, Iran.

- Project 1: Evaluation of Algebraic Attacks on Cryptosystems
- Project 2: Implementation of RFID Access Control System via AVR Microcontrollers
- Supervisor: Dr. Bahareh Akhbari

2010–2014 **Bachelor of Applied Mathematics**, *K. N. Toosi University of Technology*, Tehran, Iran.

- Supervisor: Prof. A. Reza Moghaddamfar

Research and Publications

- **Hosein Hadipour**. "Connecting Cryptanalytic Distinguishers through Divisibility". *Cryptology ePrint Archive*. Paper 2026/2471. 2026. URL: https://eprint.iacr.org/2026/2471
- Dachao Wang, **Hosein Hadipour**, and Simon Gerhalter. "On Extending Integral Distinguishers". *Cryptology ePrint Archive, Paper 2026/1402*. 2026. URL: https://eprint.iacr.org/2026/1402
- **Hosein Hadipour**, Saleh Khalaj Monfared, Jens Alich, and Jan Vorloeper. "LibFWHT: From Exact Walsh Spectra to Key Dependence in Differential-Linear Correlations". In: *IACR Trans. Symmetric Cryptol.* 2026.4 (2026)
- Shuo Peng, Akram Khalesi, Zahra Ahmadian, **Hosein Hadipour**, Jiahui He, Kai Hu, Zhongfeng Niu, Shahram Rasoolzadeh, and Meiqin Wang. "Cube and Integral Attacks on ChiLow-32". In: *IACR Trans. Symmetric Cryptol.* 2026.1 (2026), pp. 253–292. DOI: 10.46586/TOSC.V2026.I1.253-292
- Cihangir Tezcan, Gregor Leander, and **Hosein Hadipour**. "Cryptanalysis: Theory Versus Practice Correcting Cryptanalysis Results on Ascon, ChaCha, and Serpent Using GPUs". In: *IACR Trans. Symmetric Cryptol.* 2025.3 (2025), pp. 729–754. DOI: 10.46586/TOSC.V2025.I3.729-754
- **Hosein Hadipour**, Yosuke Todo, Mostafizar Rahman, Maria Eichlseder, Ravi Anand, and Takanori Isobe. "Turning Multiple Key-Dependent Attacks into Universal Attacks". *Cryptology ePrint Archive*. Paper 2025/1996. 2025. URL: https://eprint.iacr.org/2025/1996
- Chengcheng Chang, **Hosein Hadipour**, Kai Hu, Muzhou Li, and Meiqin Wang. "Mix-Basis Geometric Approach to Boomerang Distinguishers". In: *IACR Trans. Symmetric Cryptol.* 2025.3 (2025), pp. 693–728. DOI: 10.46586/TOSC.V2025.I3.693-728. URL: https://doi.org/10.46586/tosc.v2025.i3.693-728
- Debasmita Chakraborty, **Hosein Hadipour**, Anup Kumar Kundu, Mostafizar Rahman, Prathamesh Ram, Yu Sasaki, Dilip Sau, and Aman Sinha. "Breaking the Twinkle Authenticated Encryption Scheme and Analyzing Its Underlying Permutation". In: *Selected Areas in Cryptography - SAC 2025 - 32nd International Conference, Toronto, ON, Canada, August 13-15, 2025, Revised Selected Papers*. Ed. by Christina Boura, Atefeh Mashatan, and Ali Miri. Vol. 16207. Lecture Notes in Computer Science. Springer, 2025, pp. 79–109. DOI: 10.1007/978-3-032-10536-3_4
- **Hosein Hadipour**, Patrick Derbez, and Maria Eichlseder. "Revisiting Differential-Linear Attacks via a Boomerang Perspective with Application to AES, Ascon, CLEFIA, SKINNY, PRESENT, KNOT, TWINE, WARP, LBlock, Simeck, and SERPENT". in: *Advances in Cryptology - CRYPTO 2024*. Vol. 14923. LNCS. Springer, 2024, pp. 38–72. DOI: 10.1007/978-3-031-68385-5_2
- Debasmita Chakraborty, **Hosein Hadipour**, Phuong Hoa Nguyen, and Maria Eichlseder. "Finding Complete Impossible Differential Attacks on AndRX Ciphers and Efficient Distinguishers for ARX Designs". In: *IACR Trans. Symmetric Cryptol.* 2024.3 (2024), pp. 84–176. DOI: 10.46586/TOSC.V2024.I3.84-176
- **Hosein Hadipour**, Sadegh Sadeghi, and Maria Eichlseder. "Finding the Impossible: Automated Search for Full Impossible-Differential, Zero-Correlation, and Integral Attacks". In: *Advances in Cryptology - EUROCRYPT 2023*. Ed. by Carmit Hazay and Martijn Stam. Vol. 14007. LNCS. Springer, 2023, pp. 128–157. DOI: 10.1007/978-3-031-30634-1_5
- **Hosein Hadipour** and Yosuke Todo. "Cryptanalysis of QARMAv2". In: *IACR Trans. Symmetric Cryptol.* 2024.1 (2024), pp. 188–213. DOI: 10.46586/TOSC.V2024.I1.188-213

- **Hosein Hadipour**, Simon Gerhalter, Sadegh Sadeghi, and Maria Eichlseder. "Improved Search for Integral, Impossible Differential and Zero-Correlation Attacks Application to Ascon, ForkSKINNY, SKINNY, MANTIS, PRESENT and QARMAv2". In: *IACR Trans. Symmetric Cryptol.* 2024.1 (2024), pp. 234–325. DOI: 10.46586/TOSC.V2024.I1.234-325
- **Hadipour, Hosein**, Marcel Nageler, and Maria Eichlseder. "Throwing Boomerangs into Feistel Structures: Application to CLEFIA, WARP, LBlock, LBlock-s and TWINE". in: *IACR Trans. Symmetric Cryptol.* 2022.3 (2022), pp. 271–302. DOI: 10.46586/tosc.v2022.i3.271-302
- **Hosein Hadipour** and Maria Eichlseder. "Integral Cryptanalysis of WARP based on Monomial Prediction". In: *IACR Trans. Symmetric Cryptol.* 2022.2 (2022), pp. 92–112. DOI: 10.46586/tosc.v2022.i2.92-112
- **Hosein Hadipour** and Maria Eichlseder. "Autoguess: A Tool for Finding Guess-and-Determine Attacks and Key Bridges". In: *ACNS*. vol. 13269. LNCS. Springer, 2022, pp. 230–250. DOI: 10.1007/978-3-031-09234-3_12
- Hadi Soleimany, Nasour Bagheri, **Hosein Hadipour**, Prasanna Ravi, Shivam Bhasin, and Sara Mansouri. "Practical Multiple Persistent Faults Analysis". In: *IACR Trans. Cryptogr. Hardw. Embed. Syst.* 2022.1 (2022), pp. 367–390. DOI: 10.46586/tches.v2022.i1.367-390
- **Hosein Hadipour** and Nasour Bagheri. "Improved Rectangle Attacks on SKINNY and CRAFT". in: *IACR Trans. Symmetric Cryptol.* 2021.2 (2021), pp. 140–198. DOI: 10.46586/tosc.v2021.i2.140-198
- **Hosein Hadipour**, Sadegh Sadeghi, Majid M. Niknam, and Nasour Bagheri. "Comprehensive security analysis of CRAFT". in: *IACR Trans. Symmetric Cryptol.* 2019.4 (2019), pp. 290–317. DOI: 10.13154/tosc.v2019.i4.290-317

Teaching and Supervision

Mentoring	Mentored a Ph.D. student.
Bachelor's	Supervised bachelor's students.
Summer 2023, 2024	**Guest Lecturer**, *Graz University of Technology*, Graz, Austria. Delivered selected sessions of the Cryptanalysis course.
Fall 2016	**Teaching Assistant**, *University of Tehran*, Tehran, Iran. Course: A Course in Cryptography.
Fall 2014	**Teaching Assistant**, *K. N. Toosi University of Technology*, Tehran, Iran. Course: Introduction to Cryptography.

Academic Service

- **Artifact Review Chair** at FSE 2026/ToSC
- Reviewer for FSE 2027/ToSC 2026-2∼ToSC 2027-1
- Reviewer for ASIACRYPT 2026
- Reviewer for CASCADE 2026
- Reviewer for Designs, Codes and Cryptography (DCC) 2025
- Subreviewer for ASIACRYPT 2025
- Subreviewer for CRYPTO 2025
- Subreviewer for Selected Area in Cryptography (SAC) 2025
- Artifact review committee at ASIACRYPT 2024
- Subreviewer for ASIACRYPT 2024

- Subreviewer for EUROCRYPT 2024
- Subreviewer for EUROCRYPT 2023
- Subreviewer for ASIACRYPT 2023
- Subreviewer for CRYPTO 2022
- Subreviewer for ASIACRYPT 2022
- Reviewer for IET Information Security 2022
- Reviewer for Designs, Codes and Cryptography (DCC) 2022

Visits

May 2024 **Inria**, Paris, France (Visiting Professor Maria Naya-Plasencia's Lab)

May 2024 **IRISA**, Rennes, France (Visiting IRISA Lab)

May 2024 **LORIA**, Nancy, France (Visiting LORIA Lab)

Mar. 2024 **University of Hyogo**, Kobe, Japan (Visiting Professor Takanori Isobe's Lab)

Honors

2019 **Winner**, *NSUCRYPTO 2019*, October 13–21.
Competition results

2014 **Bronze Medal**, *38th National Mathematical Competition for University Students*, Kerman, Iran, May.
Competition website

2013 **Bronze Medal**, *37th National Mathematical Competition for University Students*, Semnan, Iran, May.
Competition website

Technical Skills

Programming	Python, C, C++; Java (intermediate)
Development	Git, shell scripting
Parallel/HPC	CUDA, OpenMP, Slurm
Math	SageMath, MATLAB, Maple, CoCoA
SAT	PySAT, CaDiCaL, MiniSat, CryptoMiniSat
SMT	pySMT, Z3, STP
MILP	Gurobi
CP	MiniZinc
Platforms	Linux, macOS, Windows
Typesetting	LaTeX
Earlier work	VHDL, AVR assembly; Xilinx ISE, Altium Designer, PSpice, CodeVision, Atmel Studio, Arduino, Proteus

Research Interests

- Cryptanalysis and Design of Symmetric-key Cryptographic Primitives
- Automating Cryptanalysis
- LLM-Assisted Automated Reasoning and Cryptanalysis

- Security and Reliability of AI Agents
- Side-channel and Fault Analyses
- Logic and Boolean Algebra
- Algebra and Graph Theory
- Probability Theory and Statistics
- High-Performance and Secure Implementations

Selected Talks and Presentations

Mar. 2025 **SKCAM 2025**, Rome, Italy (Invited speaker)

Dec. 2024 **ASK 2024**, TCG CREST, Kolkata, India, December 14–17 (Invited speaker)

Aug. 2024 **CRYPTO 2024**, Santa Barbara, USA, August 18–22 (Paper presentation)

Apr. 2024 **Beating Real-Time Crypto: Solutions and Analysis**, Lorentz Center, Leiden, Netherlands, April 22–26 (Invited attendee and speaker)

Mar. 2024 **FSE 2024**, Leuven, Belgium, March 25–29 (Paper presentation)

Apr. 2023 **EUROCRYPT 2023**, Lyon, France, April 23–27 (Paper presentation)

Mar. 2023 **FSE 2023**, Kobe, Japan, March 20–24 (Paper presentation)

Sept. 2022 **CHES 2022**, Leuven, Belgium, September 18–21 (Paper presentation)

Jun. 2022 **ACNS 2022**, Rome, Italy, June 20–23 (Paper presentation)

Mar. 2022 **FSE 2022**, Athens, Greece, March 20–25 (Paper presentation)

Feb. 2020 **Symmetric Cryptography in Theory and Practice**, ISC Winter School on Information Security and Cryptology (ISCwsISC 2020), Iran University of Science and Technology, February 2–3 (Speaker)

Languages

Persian	**Mother tongue**
English	**Advanced**
German	**Learning**
Chinese	**Learning**

Other Information

Memberships

2021-Present	International Association for Cryptologic Research
2012-2013	Iranian Mathematical Society
Interests	Hiking, swimming, tennis, music, and chess.